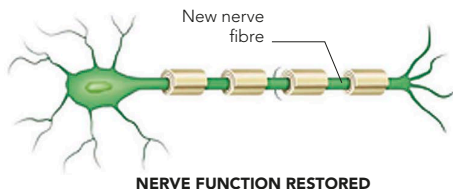
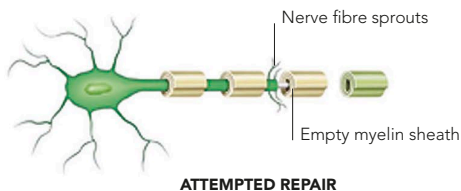
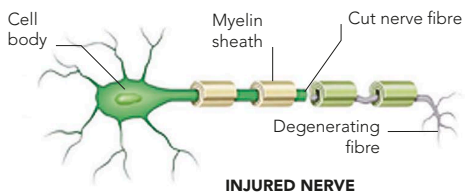


NERVE REGENERATION

Peripheral nerve fibres that have been crushed or partly cut may slowly regenerate if the cell body is undamaged. The damaged section of fibre degenerates, leaving the myelin sheath hollow. The healthy remaining fibre begins to grow along the empty sheath at a rate of 1–2mm ($\frac{1}{25}$ – $\frac{2}{25}$ in) daily. Natural regeneration is much less likely in the nerve fibres of the brain and spinal cord, where the neurons are so specialized that generally they cannot replicate themselves or recreate their highly developed connections.

REGROWTH

The stump end of a damaged nerve fibre sends out several sprout-like growths. One of these finds the empty but intact myelin sheath and grows inside it.



SUPPORT CELLS

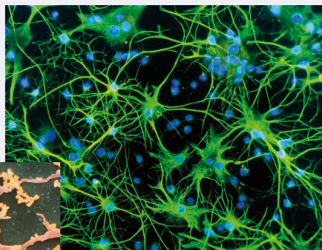
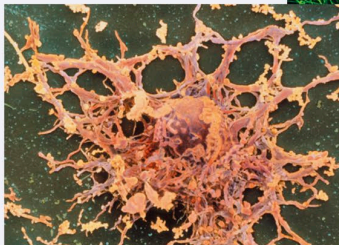
Supporting nerve cells, known as glial cells or neuroglia, protect and nourish the neurons. There are several types of glial cell.

The smallest are microglia, which destroy microorganisms, foreign particles, and cell debris from disintegrating neurons.

Ependymal cells line cavities that are filled with cerebrospinal fluid, which surrounds both the brain and spinal cord (see p.89). Other glial cells insulate the axons and dendrites or regulate the flow of cerebrospinal fluid.

ASTROCYTE

Named after their star-like appearance, they provide support and nutrition.



OLIGODENDROCYTE

These cells provide a support framework, and produce and nourish myelin sheath segments for certain axons.